ViBridge-VQA: Bridging Global and Local Visual Tokens for Parameter-Efficient Vietnamese Visual Question Answering

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Abstract. Vision–language models (VLMs) such as Vintern-1B bring strong multimodal abilities to Vietnamese, but adapting them to a new Visual Question Answering (VQA) benchmark usually means fine-tuning on many high-resolution image tiles or training a new architecture. We propose **ViBridge-VQA**, which improves Vintern-1B without updating any of its pre-trained weights. From a single image view, the frozen decoder receives *global* tokens, produced by a small trainable bridge from the `[CLS]` vector, and *local* tokens, produced by Vintern-1B's own pre-aligned projector and pooled to a budget chosen as the knee of the accuracy–cost Pareto front. On the reasoning-oriented AutoViVQA benchmark, ViBridge-VQA trains 1.35% of the parameters and reaches an F1 of 54.6, comparable to the fine-tuned Vintern-1B, while its decoder reads 36 times fewer visual tokens; a lightweight decoder LoRA raises F1 to 58.2. Reusing the pre-trained projector proves more effective than learning a new connector, and the local tokens complement decoder adaptation and transfer to three other Vietnamese VQA datasets.

Keywords: Vietnamese Visual Question Answering · Vision–Language Models · Parameter-Efficient Learning · Visual Token Budget · Generative VQA

1 Introduction

Visual Question Answering (VQA) asks a model to answer a natural-language question about an image. The task has progressed rapidly with large vision–language models (VLMs) such as BLIP-2 [12], Flamingo [1] and LLaVA [19], which connect a pre-trained vision encoder to a pre-trained language model and generate free-form answers. Vietnamese has followed this trend: the open model Vintern-1B [7] connects an InternViT-300M encoder [6] to a Qwen2.5-0.5B

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decoder [26], and benchmarks such as ViVQA [29] and the recent reasoning-oriented AutoViVQA [21] make it possible to measure progress in this low-resource setting.

Obtaining a strong model on a new Vietnamese benchmark, however, is still expensive. The common recipe fine-tunes an existing VLM: the fine-tuned Vintern-1B baseline of AutoViVQA adds LoRA adapters to the language model and splits every image into up to six high-resolution tiles plus a thumbnail, so that the decoder reads up to 1,792 visual tokens per image. The alternative is to design a task-specific architecture, as ViMoE-VQA [20] does with a sparse mixture-of-experts fusion module and a decoder trained on AutoViVQA. A cheaper option is to keep the whole VLM frozen and train only a small connector between its encoder and decoder. Existing connectors follow two designs, each with its own weakness. A *global-only bridge* maps the global image vector to a few tokens; it is cheap, but it loses local details such as small objects, printed text and spatial relations. A *learned patch compressor*, such as a Q-Former, keeps these details, but it has to learn a new mapping from visual features to the decoder, which is difficult with limited Vietnamese data.

A third option is already present in Vintern-1B. Its projector was pre-trained to map image regions into the input space of the decoder, so this alignment already exists and does not need to be learned again. At the same time, the model discards the global `[CLS]` vector of its encoder, which summarises the whole scene. We therefore study whether, with limited labelled data, reusing the pre-trained alignment of a VLM is more effective than learning a new connector, and how many global and local tokens the frozen decoder needs.

To this end, we propose **ViBridge-VQA**, which gives the frozen decoder of Vintern-1B two complementary kinds of visual tokens from a single image view. *Local* tokens reuse the frozen projector of Vintern-1B, average-pooled to a chosen budget and placed in its original image slot, so the decoder receives region features in the representation it was trained to read. *Global* tokens are produced from the `[CLS]` vector by a small trainable bridge and add the scene-level information that Vintern-1B ignores. Only this bridge is trained. Because every additional token increases inference cost, we treat the number of global and local tokens as a budget and select it as the knee of the accuracy–cost Pareto front. As a result, ViBridge-VQA performs on par with the published fine-tuned Vintern-1B without updating any of its weights, while its decoder reads far fewer visual tokens.

In summary, our main contributions are as follows:

- We propose **ViBridge-VQA**, a global–local bridge that improves a frozen Vietnamese VLM by combining trainable global tokens with local tokens from its own pre-trained projector, and that can be combined with a lightweight decoder LoRA.
- We treat the number of visual tokens as an explicit **accuracy–cost budget** and select it on the validation split as the knee of the Pareto front.
- We **analyse where the gains come from**: local tokens matter far more than global ones, reusing the pre-trained projector works better than learning

a new connector, local tokens complement decoder adaptation, and they transfer to other Vietnamese VQA datasets.

2 Related Work

Early VQA systems classified over a fixed set of answers, whereas current generative VLMs produce free-form answers with an autoregressive language model [12,1,19]. For Vietnamese, available VQA datasets include ViVQA [29], the synthetic ViCLEVR [31], the book-cover ViOCRVQA [25] and the scene-text ViTextVQA [23]. AutoViVQA [21] contributes 37,077 questions on 19,411 COCO images [17], generated by large language models, filtered by an ensemble of validators and annotated with reasoning categories, with an emphasis on relational, causal and multi-step reasoning. On the modelling side, BARTPhoBEiT [30] and AViVQA-TranConI [22] combine pre-trained visual and Vietnamese language representations, Vintern-1B [7] brings an instruction-tuned VLM to Vietnamese, and ViMoE-VQA [20] routes multimodal tokens to sparse experts for generative VQA on AutoViVQA. AutoViVQA and ViMoE-VQA are earlier work of our group; here we use the AutoViVQA data but no component of ViMoE-VQA, and the method, the image-disjoint split and all our results are new. These approaches either train a new model or fine-tune an existing VLM on multi-tile inputs; in contrast, we keep every pre-trained weight of Vintern-1B frozen and use a single image view.

Keeping the vision encoder and the language model frozen and training only a connector is a common way to build VLMs efficiently, and connectors differ mainly in what they must learn. Frozen [32] learns a visual prefix from a global image representation. The Q-Former of BLIP-2 [12] learns a set of queries that compress the patch features, which preserves details but requires a large module (188M parameters) pre-trained on millions of image–text pairs. LLaVA [19,18] projects every patch token with a small MLP, which preserves locality but passes hundreds of tokens to the decoder; Honeybee [4] proposes locality-preserving abstractors, and InternVL [6], on which Vintern-1B is built, merges neighbouring patches before its MLP projector. In all of these models, the mapping from patches to the decoder is learned during large-scale pre-training. ViBridge-VQA differs in two respects: it does not learn this local mapping again but reuses the projector already pre-trained in the VLM, and it trains only a small global bridge (12.9M parameters) on about 26k task questions.

The number of visual tokens largely determines the cost of the decoder. Inference-optimal scaling of VLMs [13] shows that a few tokens can suffice for many tasks, and Matryoshka multimodal models [3] train a model to work at several nested pooled token budgets and show that the required budget depends on the task. Other work reduces the visual tokens directly: DeCo [34] replaces the Q-Former with adaptive average pooling of the patch tokens and reports better results, TokenPacker [14] learns a compact projector, and LLaVA-PruMerge [28] and FastV [5] prune or merge visual tokens of an existing model. Like DeCo, we rely on simple average pooling, but we apply it to the output of a projector that

is already pre-trained, keep the whole VLM frozen, and choose among budgets with explicit criteria, namely the knee of the Pareto front [27] and TOPSIS [11]. Parameter-efficient methods such as adapters [9], prefix tuning [15] and LoRA [10] reduce the cost of adaptation from the other side, by training a small number of additional weights in the decoder. We use decoder LoRA both as a baseline and as a complement, and ask whether a small training budget is better spent on the visual tokens or on adapting the decoder.

3 Proposed Method

We formulate VQA as conditional answer generation: given an image I and a Vietnamese question q, the model generates a short free-form answer $y = (y_1, \ldots, y_T)$. ViBridge-VQA builds on Vintern-1B [7] and keeps all of its pre-trained weights frozen. Instead of adapting the model, it changes the visual input of its decoder.

3.1 Overall Architecture

As illustrated in Fig. 1, ViBridge-VQA keeps the three components of Vintern-1B: the InternViT-300M vision encoder (24 layers, width 1024), its two-layer MLP projector P, and the Qwen2.5-0.5B-Instruct language decoder (24 layers, width 896). Each image is resized to one 336×336 view (24×24 patches of 14×14 pixels); InternViT, pre-trained at 448×448, bicubically interpolates its position embeddings to this resolution, which roughly halves the encoder cost (191 instead of 362 GMACs per view). The encoder returns one vector per patch and a global `[CLS]` vector $v \in \mathbb{R}^{1024}$. Vintern-1B uses only the patch vectors: it merges every 2×2 block of neighbouring patches into one 4096-dimensional vector (pixel shuffle) and maps each merged vector into the decoder input space with P, which was pre-trained to align the two backbones. At our resolution this yields a 12×12 grid of 144 region tokens. ViBridge-VQA uses both outputs of the encoder.

3.2 Global Tokens

The `[CLS]` vector summarises the whole scene, but Vintern-1B never passes it to the decoder. A small trainable bridge g_ϕ expands it into g tokens with two linear heads:

$$g_\phi(v) = \left[\, W_a v + b_a;\ \mathrm{reshape}_{(g-1)\times 896}(W_e v + b_e) \,\right] \in \mathbb{R}^{g \times 896}. \tag{1}$$

where $W_a \in \mathbb{R}^{896\times 1024}$ and $W_e \in \mathbb{R}^{(g-1)896\times 1024}$. The first head produces an *anchor* token, and the second produces $g-1$ complementary tokens that present the global content to the decoder from different learned directions. The bridge has $1025 \cdot 896 \cdot g$ parameters, i.e., 12,857,600 for $g = 14$, and its g tokens are placed before the prompt. We consider $g \in \{8, 14\}$: $g = 8$ is our Global-only model, as it matches the eight learned queries of the Tile-Attention and light

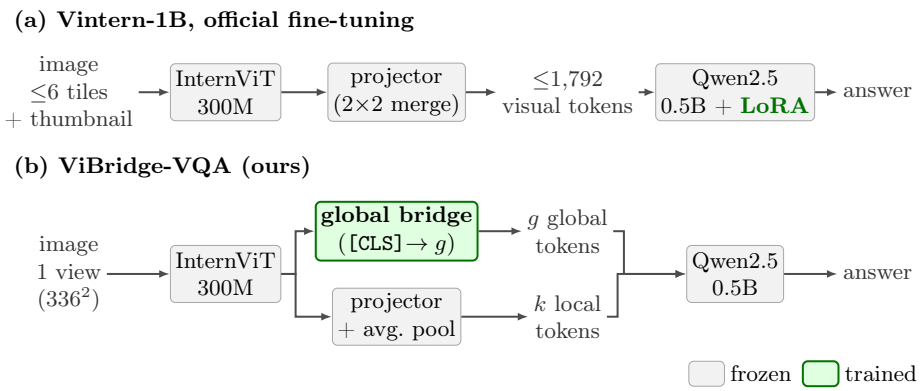


Fig. 1: Overview of (a) the official fine-tuning of Vintern-1B, which encodes up to seven views, and (b) ViBridge-VQA, which encodes one view and passes g global and k local tokens to the frozen decoder (selected: $g = 14$, $k = 36$).

Q-Former bridges it is compared with (Section 4.4), and $g = 14$ is where F1 stopped increasing when the Global-only bridge was varied from 4 to 20 tokens (Section 4.5). The final pair (g, k) is selected with the criterion of Section 3.5.

3.3 Local Tokens

To give the decoder region-level detail, ViBridge-VQA also passes local tokens, taken directly from the 144 region tokens of the frozen projector P. To control their number, we average-pool the 12×12 grid to an $s \times s$ grid, so that $k = s^2$ with $s \in \{1, 3, 6, 12\}$: $k = 144$ keeps every region token, and $k = 36$ averages each 2×2 block. The pooled tokens replace the embeddings of k image-context placeholders inside Vintern-1B's own image slot, `<img>` $\cdots$ `</img>`, in the prompt. Because they come from the pre-trained projector rather than from a newly learned projection of the patches, the local tokens stay in the representation that the decoder was trained to read, and no parameters are needed to produce them.

3.4 Training Objective

Only the bridge parameters ϕ are trained; the vision encoder, the projector and the decoder remain frozen, and no adapter is added to the decoder. We minimise the cross-entropy of the answer tokens:

$$\mathcal{L} = -\sum_{t=1}^{T} \log p\big(y_t \mid y_{<t},\, g_\phi(v),\, \mathrm{pool}_s(P(x)),\, q\big), \tag{2}$$

where x denotes the merged patch vectors. ViBridge-VQA can also be combined with decoder adaptation: in that variant, a rank-16 LoRA [10] on the query, key, value and output projections of every decoder layer is trained jointly with the bridge, which adds 2,162,688 parameters. Throughout the paper, percentages of

trained parameters are relative to all parameters of the model, i.e., the frozen components of Vintern-1B (about 0.94B, including the vision encoder, projector, embeddings and output layer) plus the trained modules. With $g = 14$, the bridge alone trains 12.86M parameters (1.35%), and the bridge with LoRA 15.02M (1.57%).

3.5 Choosing the Token Budget

The cost of the decoder grows with the number of visual tokens $g + k$, so we treat the pair (g, k) as a budget to be chosen. In our experiments, F1 increases with every added token among the configurations with local tokens, so all of them lie on the accuracy–token Pareto front and the front alone does not single one out. We therefore select its *knee* [27], the point after which additional tokens bring diminishing returns. After normalising cost c and accuracy a to $[0, 1]$ over the front, the knee is the configuration that maximises $\hat{a} - \hat{c}$, i.e., the one lying furthest above the line joining the cheapest and the most accurate configuration. The knee is computed on the validation split only, and we check the choice with the multi-criteria method TOPSIS [11], also on validation. Test results are computed for every configuration only to be reported; no choice is based on them.

4 Experiments

4.1 Dataset and Evaluation Metrics

Dataset. We use AutoViVQA [21], a Vietnamese VQA benchmark on MS COCO images [17] with five short free-form reference answers and a reasoning type per question. We split it by image, so that no test image is seen in training: after removing 273 duplicated (image, question) pairs and 97 questions without one of the eight reasoning labels, the 36,707 questions are divided 70/15/15 by image (seed 42) into 25,776 / 5,463 / 5,468 questions on 13,579 / 2,909 / 2,910 images. This split is stricter: it lowers the F1 of Global-only by about 0.5 points compared with a split that does not group questions by image.

Metrics. Following AutoViVQA and ViMoE-VQA, we report Accuracy (exact match), word-level Precision, Recall and F1, BLEU [24], ROUGE-L [16], METEOR [2] and CIDEr [33] against the five reference answers.

4.2 Experimental Setup

Implementation details. The vision encoder, the projector and the decoder are taken from Vintern-1B-v3.5 [7] and kept frozen. We use AdamW (learning rate 2×10^{-4}, effective batch size 8) and train on the first reference answer: two epochs for ViBridge-VQA, after which neither the validation loss nor CIDEr improves, and one joint epoch for the variants with a decoder LoRA; Global-only + LoRA

Table 1: Performance comparison on the AutoViVQA test set. Baseline results are taken from AutoViVQA [21] and ViMoE-VQA [20]; ours are the mean ± std over three seeds. Best results per column are in bold. All values ×100.

	Model	Accuracy	Precision	Recall	F1	BLEU	ROUGE-L	METEOR	CIDEr
Fine-tuned	ViT5_ViT	7.97	46.84	50.33	48.52	4.13	46.89	31.02	72.68
	BARTPhoBEiT	8.81	45.30	46.48	45.88	4.33	44.83	24.57	**188.96**
	Vintern-1B	13.01	52.47	55.12	53.76	6.11	51.93	35.25	72.84
	ViMoE-VQA	9.55	**62.89**	58.60	**60.69**	12.54	47.12	39.10	88.76
Zero-shot	Vintern-1B	0.12	17.52	19.87	17.55	1.91	25.84	23.93	8.54
	LLaMA 3.2	0.36	23.96	73.71	36.16	3.62	36.11	30.01	62.84
	Gemini 2.0 F.	0.55	27.20	74.10	39.79	4.41	39.60	31.72	74.42
	Gemini 2.5 F.	0.22	24.43	**76.66**	24.75	0.39	37.27	31.22	71.90
	GPT-5	10.84	47.20	55.20	50.89	6.07	47.30	33.34	84.20
Ours	ViBridge-VQA (k=36)	11.75 ±0.21	55.36 ±0.21	56.56 ±0.21	54.56 ±0.19	18.52 ±0.18	52.64 ±0.19	45.53 ±0.33	110.58 ±0.95
	ViBridge-VQA (k=36) + LoRA	**14.67** ±0.30	59.02 ±0.12	59.96 ±0.33	58.17 ±0.19	**22.53** ±0.11	**56.25** ±0.22	**49.32** ±0.37	122.88 ±0.93

is also trained for three epochs, its best setting (+1.3 validation F1). Decoding is greedy (at most 50 new tokens). Each configuration is trained with three seeds (42, 123, 3407), and we report the mean ± std. Every model fits on a single 16 GB NVIDIA T4. The token budget is selected on validation; the test split is used only for the final report.

Baselines. We compare with (1) multimodal models fine-tuned on AutoViVQA: ViT5_ViT, BARTPhoBEiT [30], Vintern-1B and ViMoE-VQA [20]; (2) zero-shot models: Vintern-1B, LLaMA 3.2, Gemini 2.0 and 2.5 Flash, and GPT-5; and (3) *Global-only*, the Multi-Token bridge of Eq. (1) with $g = 8$ and no local tokens, without and with a decoder LoRA (*Global-only + LoRA*). Results of groups (1) and (2) are taken from the AutoViVQA and ViMoE-VQA papers; the fine-tuned Vintern-1B follows the official recipe (up to six 448 × 448 tiles plus a thumbnail, rank-16 LoRA on the language model for one epoch). Our models are *ViBridge-VQA (k=36)*, the selected budget ($g = 14$, $k = 36$), and *ViBridge-VQA (k=36) + LoRA*.

4.3 Main Results

Table 1 compares ViBridge-VQA with the baselines. Without updating any weight of Vintern-1B, ViBridge-VQA raises its F1 from 17.55 (zero-shot) to 54.56, comparable to the fine-tuned Vintern-1B (53.76), while its decoder reads 50 instead of 1,792 visual tokens and only 1.35% of the parameters are trained; only its Accuracy is lower (11.75 versus 13.01). A decoder LoRA (1.57% of the parameters trained) raises F1 to 58.17 (+3.6) and gives the best Accuracy, BLEU, ROUGE-L and METEOR. ViMoE-VQA, whose fusion module and decoder are trained on AutoViVQA, keeps the best Precision and F1. With the same one-epoch LoRA, ViBridge-VQA is 4.7 validation F1 above Global-only (58.21 versus 53.47; Table 2). Results are stable across seeds: on test, the std is at most 0.19 for F1 and 0.95 for CIDEr, and test F1 is within 0.6 of validation F1.

Table 2: Comparison of visual-token sources on the validation set (mean ± std over three seeds, four for Multi-Token). Params: trained bridge parameters. Best results per block are in bold. All values ×100.

Visual tokens from	Tokens	Params	F1	BLEU	ROUGE-L	METEOR	CIDEr
Bridge only (two epochs)							
Residual	1	4.86M	46.02 ±.36	12.98 ±.15	44.23 ±.38	36.86 ±.31	86.48 ±.77
Multi-Token	8	7.35M	50.74 ±.17	16.88 ±.23	49.03 ±.12	41.38 ±.21	99.07 ±.47
Tile-Attention	8	4.14M	45.14 ±.92	12.34 ±.32	43.28 ±.95	36.11 ±.73	84.14 ±1.40
Light Q-Former	8	27.6M	47.06 ±.66	13.99 ±.33	45.37 ±.63	37.57 ±.60	88.07 ±1.89
Full Q-Former	16	69.4M	46.84 ±.47	13.76 ±.36	45.10 ±.41	37.43 ±.34	87.74 ±1.24
ViBridge-VQA (k=36)	50	12.86M	**54.96** ±.28	**18.73** ±.23	**53.10** ±.27	**46.01** ±.42	**113.29** ±1.01
+ decoder LoRA (one joint epoch, 2.16M parameters)							
Residual	1	4.86M	52.69 ±.14	18.94 ±.36	51.04 ±.13	43.33 ±.33	104.25 ±.93
Multi-Token	8	7.35M	53.47 ±.17	19.72 ±.09	51.79 ±.15	44.11 ±.23	106.68 ±.69
Tile-Attention	8	4.14M	52.80 ±.06	19.27 ±.16	51.16 ±.06	43.41 ±.05	104.81 ±.14
Light Q-Former	8	27.6M	53.21 ±.14	19.92 ±.31	51.54 ±.13	43.77 ±.19	106.26 ±.73
Full Q-Former	16	69.4M	53.29 ±.29	19.57 ±.25	51.62 ±.30	43.96 ±.21	106.50 ±1.04
ViBridge-VQA (k=36)	50	12.86M	**58.21** ±.16	**22.27** ±.17	**56.42** ±.14	**49.55** ±.06	**124.88** ±.75

4.4 Source of the Visual Tokens

Table 2 compares three sources of visual tokens: the `[CLS]` vector (*Residual*, one token, and *Multi-Token*, Eq. (1)), learned compressions of the patches (*Tile-Attention* with eight queries, light and full Q-Former [12]), and our local tokens from the pre-trained projector.

Reusing the projector works best. Without decoder adaptation, the learned compressions stay 3.7–5.6 F1 below Multi-Token despite up to 9.4× more parameters, whereas adding 36 or 144 local tokens to the same bridge ($g = 8$) raises F1 by 3.6 and 5.4 (Table 3). Training Global-only to imitate the projector does not close the gap: aligning its mean output token with the projector's (cosine) or its answer distribution with the one obtained from the projector's tokens (KL) gives 50.35 and 50.46 F1, against 50.74.

Local tokens and LoRA are complementary. With a decoder LoRA, every bridge without local tokens reaches 52.69–53.47 F1, still below ViBridge-VQA with a frozen decoder (54.96). The same LoRA adds 3.3 F1 to ViBridge-VQA, more than the 2.7 it adds to Multi-Token.

4.5 Visual Token Budget

Local tokens matter more than global tokens. At $g = 14$, going from 1 to 144 local tokens adds 4.9 F1 and 19.1 CIDEr (Table 3, Fig. 2a), whereas going from 8 to 14 global tokens adds 0.2–0.8 F1 at a fixed k. The gain per token falls from 0.07 F1 between 23 and 50 tokens to 0.015 between 50 and 158 tokens.

Selected budget. Since F1 grows with every added token, all configurations with local tokens lie on the accuracy–token Pareto front. Its knee (Section 3.5) is

Table 3: Effect of the token budget on the validation set (mean ± std over three seeds, four for $g = 8$, $k = 0$). Time: minutes to evaluate validation and test on one T4. The selected budget is in bold. All values ×100.

g	k	Tokens	Time	F1	BLEU	ROUGE-L	METEOR	CIDEr
8	0	8	–	50.74 ±.17	16.88 ±.23	49.03 ±.12	41.38 ±.21	99.07 ±.47
14	0	14	–	51.56 ±.08	17.76 ±.01	49.86 ±.05	42.14 ±.17	101.18 ±.52
8	1	9	57.0	51.03 ±.12	16.54 ±.27	49.21 ±.13	41.37 ±.11	99.27 ±.58
8	9	17	59.2	52.82 ±.11	17.58 ±.03	50.96 ±.05	43.56 ±.20	106.11 ±.55
8	36	44	60.6	54.32 ±.41	18.22 ±.39	52.48 ±.43	45.31 ±.42	111.88 ±1.43
8	144	152	71.2	56.15 ±.08	19.82 ±.00	54.21 ±.06	47.51 ±.05	119.05 ±.25
14	1	15	58.8	51.67 ±.25	17.30 ±.36	49.93 ±.24	42.36 ±.19	101.66 ±1.11
14	9	23	57.7	52.98 ±.08	17.83 ±.23	51.17 ±.10	43.71 ±.13	106.02 ±.80
14	**36**	**50**	**60.0**	**54.96** ±.28	**18.73** ±.23	**53.10** ±.27	**46.01** ±.42	**113.29** ±1.01
14	144	158	70.2	56.61 ±.04	20.18 ±.13	54.70 ±.05	48.00 ±.05	120.71 ±.12

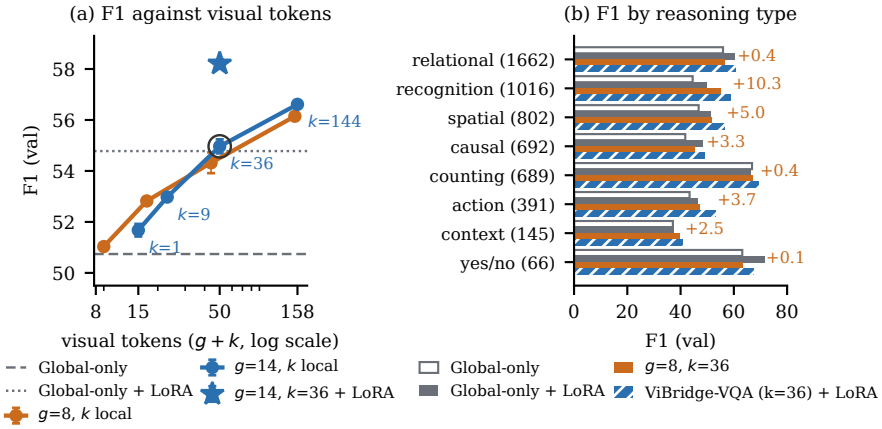


Fig. 2: (a) Validation F1 against the number of visual tokens (circle: selected budget). (b) Validation F1 by reasoning type (number of questions); numbers: gain of 36 local tokens at $g = 8$.

$g = 14$, $k = 36$ for every metric, and also with evaluation time as the cost (times differ by only minutes below $k = 144$). TOPSIS with equal weights and the knee on the test split (a post-hoc check) give the same budget. $k = 144$ is 1.7 F1 higher, but $k = 36$ obtains 72% of its gain over Global-only with 32% of its tokens. Against the 14-token Global-only bridge ($g = 14$, $k = 0$), the 36 local tokens add 3.4 F1 (51.56 to 54.96), 67% of the gain of $k = 144$.

Number of global tokens. Varying the Multi-Token bridge from $g = 4$ to $g = 20$ (three seeds; two for $g = 20$), validation F1 rises from 49.0 ($g = 4$) to 50.7 ($g = 8$) and 51.6 ($g = 14$), then stays at 51.2–51.6 for $g = 16$–20.



Fig. 3: Qualitative examples from the validation set. GT: one of the five reference answers; Ours: ViBridge-VQA (k=36); Ours+LoRA: ViBridge-VQA (k=36) + LoRA.

Computational cost. The decoder reads 50 visual tokens instead of the 1,792 of the official tiling (seven views for every AutoViVQA test image), and the encoder processes one 336×336 view (191 GMACs) instead of seven 448×448 views (362 GMACs each), 13 times less computation. Within our budgets, $k = 144$ instead of 36 makes evaluation about 17% slower.

4.6 Analysis

Reasoning types. Local tokens help most when the answer depends on visible content (Fig. 2b): at $g = 8$, 36 local tokens add 10.3 F1 on recognition, 5.0 on spatial and 3.7 on action questions, where this frozen-decoder model also beats Global-only + LoRA. Decoder adaptation helps more on causal and relational questions, which have the longest answers (5.0 and 5.6 words), and on yes/no questions. ViBridge-VQA (k=36) + LoRA is the best of the four models on seven of the eight types, all but yes/no.

Error analysis. Our answers are as long as the references (4.3 words on average), and 77–81% of them partially overlap the closest reference. Compared with Global-only, ViBridge-VQA raises exact matches (F1 = 1) from 9.7% to 13.3% and lowers answers sharing no word with any reference from 9.6% to 8.0% (6.8% with LoRA).

Table 4: Cross-dataset results (mean ± std over three samples of about 1,000 test questions, one training seed). Best results per dataset are in bold. All values ×100.

Model	F1	BLEU	ROUGE-L	METEOR	CIDEr
ViTextVQA					
Vintern-1B, zero-shot	**32.01** ±.31	**4.94** ±.18	**20.89** ±.25	**26.50** ±.54	**195.62** ±2.35
Global-only + LoRA	3.96 ±.09	0.15 ±.01	3.74 ±.12	2.61 ±.16	15.64 ±.92
ViBridge-VQA (k=36) + LoRA	15.39 ±.39	1.03 ±.15	14.41 ±.33	11.17 ±.26	70.07 ±1.33
ViVQA-X					
Vintern-1B, zero-shot	14.52 ±.47	0.06 ±.01	8.98 ±.32	11.96 ±.41	50.13 ±2.83
Global-only + LoRA	15.89 ±.65	0.18 ±.02	11.94 ±.49	9.08 ±.57	42.12 ±2.92
ViBridge-VQA (k=36) + LoRA	**27.64** ±.11	**0.33** ±.05	**21.98** ±.13	**19.24** ±.14	**95.29** ±1.51
ViVQA					
Vintern-1B, zero-shot	23.90 ±.24	0.27 ±.02	15.78 ±.25	17.84 ±.19	78.63 ±2.07
Global-only + LoRA	31.48 ±1.15	0.25 ±.09	31.48 ±1.16	21.92 ±.82	84.94 ±2.87
ViBridge-VQA (k=36) + LoRA	**44.76** ±1.21	**0.63** ±.02	**44.74** ±1.22	**36.98** ±1.28	**183.60** ±7.17

Qualitative examples. In Fig. 3, ViBridge-VQA already finds the relevant content (the driver's phone, the helmets, the football match, the bird feeder) but answers vaguely; the decoder LoRA makes the answer precise, e.g., the driver is using *the phone* and the helmets are worn *to protect their heads*.

4.7 Generalisation to Other Datasets

Table 4 evaluates the models on three Vietnamese VQA datasets not used for training: the scene-text ViTextVQA [23] and the general-domain ViVQA-X [8] and ViVQA [29], both on COCO images (about 10% of the sampled questions concern images of our training split, with other questions). ViBridge-VQA (k=36) + LoRA outperforms Global-only + LoRA on all three datasets and the zero-shot Vintern-1B on the general-domain ones (F1 27.6 versus 14.5 on ViVQA-X, 44.8 versus 23.9 on ViVQA). On scene text, Vintern-1B, which reads up to seven 448×448 views, stays ahead (32.0 versus 15.4 F1). Averaged over the three datasets, our model reaches 29.3 F1, against 23.5 for Vintern-1B.

5 Conclusion

In this work, we introduced **ViBridge-VQA**, which improves the Vietnamese VLM Vintern-1B for generative VQA without updating any of its pre-trained weights. Our results show that, for a VLM whose projector is already aligned with its decoder, reusing this projector with a small number of pooled local tokens is more effective than learning a new connector from limited data, that the number of tokens can be chosen with an explicit accuracy–cost criterion, and that the local tokens complement decoder adaptation rather than replace it.

Our study also has limitations. We train on a single dataset, AutoViVQA, and the task-specific ViMoE-VQA still reaches a higher F1. Without decoder adaptation, the exact-match accuracy of ViBridge-VQA stays below that of the fine-tuned Vintern-1B, and with 50 visual tokens it reads scene text much worse

than Vintern-1B; accuracy had not saturated at the 144 local tokens available at our resolution.

Overall, an existing Vietnamese VLM can be improved substantially by choosing which visual tokens its frozen decoder receives, and how many. The code and the image-disjoint split are available at `https://github.com/nguyennn263/modular-vlm-finetune`.

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